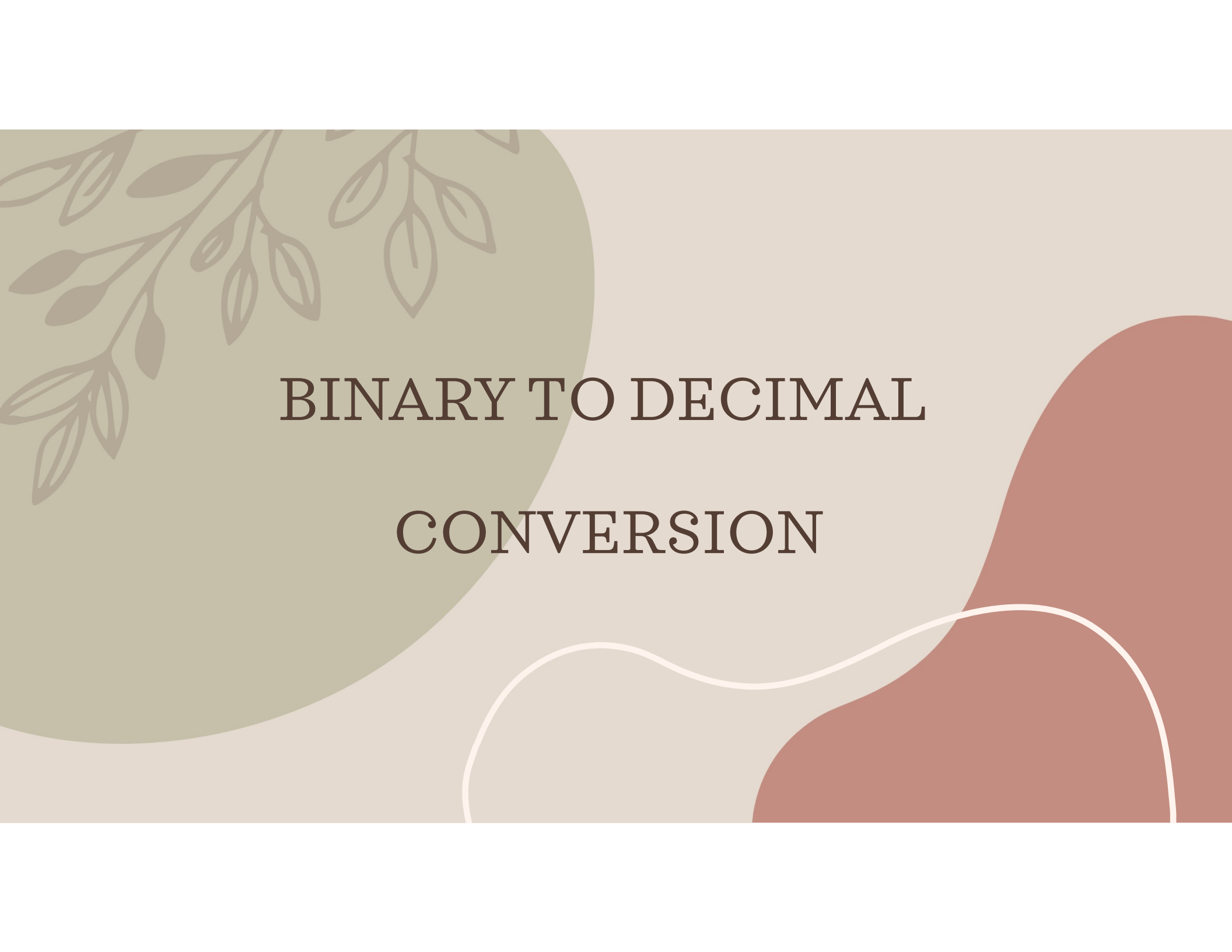




BINARY TO DECIMAL CONVERSION

BINARY TO DECIMAL CONVERSION – CONCEPT

1. In the binary number system, we use only two digits: 0 and 1.
2. Each position in a binary number has a weight, which is a power of 2.

Starting from the rightmost bit:

$$2^5=32 \quad 2^4=16 \quad 2^3=8 \quad 2^2=4 \quad 2^1=2 \quad 2^0=1$$

1010 ₂	1	0	1	0
	2^3	2^2	2^1	2^0

To convert a binary number to decimal:

1. Write the powers of 2 below each bit.

1. Multiply each bit by its corresponding power of 2.

1. Add all the results.

Example 1

Binary Number: 1010_2

Bit	1	0	1	0
Power	2^3	2^2	2^1	2^0

$$1 \times 8 = 8 \quad 0 \times 4 = 0 \quad 1 \times 2 = 2 \quad 0 \times 1 = 0$$

$$8 + 0 + 2 + 0 = 10_{10}$$

$$1010_2 = 10_{10}$$

Example 2

Binary Number: 11001_2

Bit	1	1	0	0	1
Power	2^4	2^3	2^2	2^1	2^0

$$1 \times 16 = 16$$

$$1 \times 8 = 8$$

$$0 \times 4 = 0$$

$$0 \times 2 = 0$$

$$1 \times 1 = 1$$

$$16 + 8 + 1 = 25$$

$$11001_2 = 25_{10}$$

Example 4

Convert $(101.101)_2$

101.101_2

For the integer part, we use positive powers of 2.
For the fractional part, we use negative powers of 2.

Bit	1	0	1	.	1	0	1
Weight	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}

$$1 \times 2^2 = 4$$

$$1 \times 2^0 = 1$$

$$1 \times 2^{-1} = 0.5$$

$$1 \times 2^{-3} = 0.125$$

$$\text{Integer part} = 4 + 0 + 1 = 5$$

$$\text{Fractional part} = 0.5 + 0 + 0.125 = 0.625$$

$$5 + 0.625 = 5.625$$

$$(101.101)_2 = (5.625)_{10}$$

Practice Questions

1. $101.101_2 = ?_{10}$
2. $110.011_2 = ?_{10}$
2. $11010_2 = ?_{10}$
4. $101101_2 = ?_{10}$



thank you

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